

# AWS Well-Architected Review: Designed for Efficient Cloud-Native Development

The AWS Well-Architected Review (WAR) is a systematic approach to evaluating systems on the AWS cloud against a set of architectural best practices.

This review is designed to provide developers with a clear understanding of their applications' cloud performance, identifying areas of improvement to ensure the highest quality deliverables. Let's see how it works.



**W**ithin the dynamic domain of cloud computing, where agility and scalability are of the utmost importance, it is critical to prioritise quality and efficiency for cloud applications. The implementation of AWS (Amazon Web Services) Well-Architected Review (WAR) has provided organisations with a systematic framework through which they can assess and enhance their cloud architectures. In addition to improving the calibre of cloud application development, these evaluations are instrumental in maximising the cost-effectiveness of cloud deployment. They play a crucial role throughout the software development life cycle (SDLC) such as:

- **Planning and analysis:** During the initial stages of the SDLC, the review provides guidance on making informed decisions about architecture design, helping to plan efficient and scalable systems.
- **Design and development:** The review's best practices guide developers in designing robust, efficient, and secure applications. This helps in making design choices

that ensure high performance and cost-efficiency.

- **Testing:** The review aids in identifying potential issues and vulnerabilities in the system, enabling developers to address them during the testing phase.
- **Deployment and maintenance:** Post-deployment, the review serves as a continuous improvement tool, helping to identify areas for refinement and optimisation.

## Six pillars of the AWS well-architected framework

AWS Well-Architected Review is based on a well-architected framework, which is built on six pillars.

- **Operational excellence:** The framework focuses on running and monitoring systems to deliver business value as well as continually improving processes and procedures.
- **Security:** Concentrates on protecting information and systems.
- **Reliability:** Emphasises the ability to prevent and quickly recover from failures to meet business and customer demand.
- **Performance efficiency:** Focuses on using IT and computing resources efficiently.
- **Cost optimisation:** Concentrates on avoiding unnecessary costs.
- **Sustainability:** AWS has introduced sustainability as the sixth pillar to focus on reusable energy and carbon footprint reduction. A few best practices prescribed by the AWS well-architected framework to enable good quality application delivery are listed below.
  - **Perform operations as code** – Implement infrastructure services as configuration and code so that these can be altered or ported to different environment and tuning requirements in future.
  - **Annotate documentation** – This helps future users understand code and configuration better.
  - **Make frequent, small, reversible changes** – This is to avoid confusing, lengthy and non-debuggable code during development.
  - **Refine operations procedures frequently** – This helps to optimise